From installation failure to a working simulation

QoS Performance Analysis of 5G Non-Terrestrial Networks

ns-3.46 / Ubuntu 24.04 / GitHub Codespaces

A2 — Setup & Traffic Model

What A2 needed to demonstrate

- **Working simulator**
  Install, configure and run ns-3.
- **Traffic generation**
  Implement a first stochastic packet source.
- **Evidence and analysis**
  Export CSV and PCAP; visualize PDF and CDF.
- **Reproducibility**
  Document problems, source code and results in GitHub.

First approach: local WSL2 installation

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> wsl --install
Herunterladen: Windows-Subsystem für Linux 3.0.1
Installation: Windows-Subsystem für Linux 3.0.1
Windows-Subsystem für Linux 3.0.1 wurde installiert.
Optionale Windows-Komponente wird installiert: VirtualMachinePlatform

Tool zur Imageverwaltung für die Bereitstellung
Version: 10.0.26100.8972

Abbildversion: 10.0.26200.9457
```

Windows PowerShell

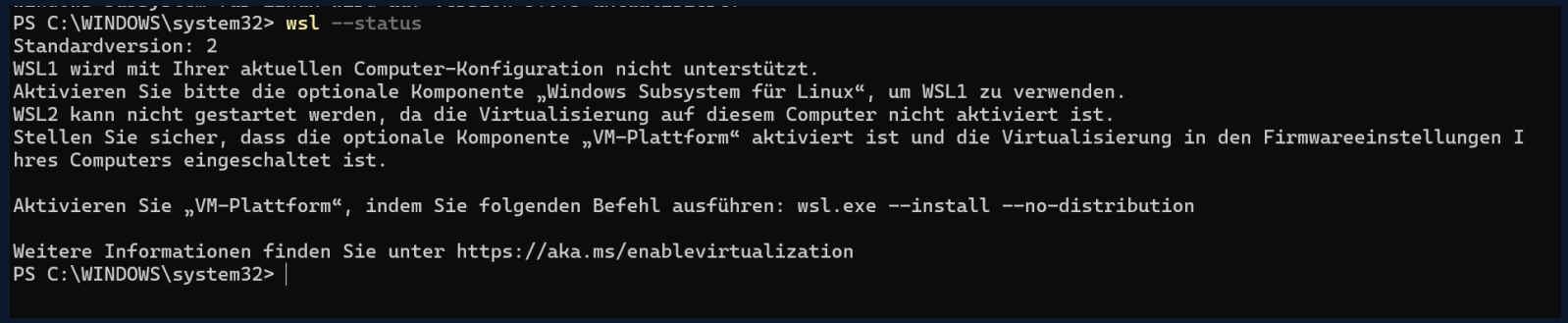
wsl --install

WSL 3.0.1 installed.

VirtualMachinePlatform enabled.

Restart requested.

The unexpected setback — and diagnosis



Windows recovery was required

After repair: hardware virtualization enabled, but required Windows features disabled.

Root cause of the boot issue remained unconfirmed.

Recovery: move the development environment to the cloud

```
@hueberpatrick ➜ /workspaces/MWN-2026
@hueberpatrick ➜ /workspaces/MWN-2026
@hueberpatrick ➜ /workspaces/MWN-2026-F11502008-hueberpatrick (main) $ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.5 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
@hueberpatrick ➜ /workspaces/MWN-2026-F11502008-hueberpatrick (main) $
```

GitHub Codespaces

Ubuntu 24.04.5 LTS

Browser-based Linux terminal

No further Windows configuration

Decision: prioritize a reproducible working environment over further local troubleshooting.

A second issue: Python compatibility

```
@hueberpatrick → ~/ns-3 (ea50b72) $ cd /workspaces/MWN-2026-F11502008-hueberpatrick

git add A2-ns3-setup

git commit -m "Add ns-3 traffic simulation and analysis results"

git push
[main 28e283a] Add ns-3 traffic simulation and analysis results
 9 files changed, 312 insertions(+)
 create mode 100644 A2-ns3-setup/results/interarrival_cdf.png
 create mode 100644 A2-ns3-setup/results/interarrival_pdf.png
 create mode 100644 A2-ns3-setup/results/ntn-traffic-0-0.pcap
 create mode 100644 A2-ns3-setup/results/ntn-traffic-1-0.pcap
 create mode 100644 A2-ns3-setup/results/ntn-traffic.csv
 create mode 100644 A2-ns3-setup/results/packet_size_cdf.png
 create mode 100644 A2-ns3-setup/results/packet_size_pdf.png
 create mode 100644 A2-ns3-setup/src/analyze_traffic.py
 create mode 100644 A2-ns3-setup/src/ntn-traffic.cc
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 2 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (14/14), 151.05 KiB | 11.62 MiB/s, done.
Total 14 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/hueberpatrick/MWN-2026-F11502008-hueberpatrick
   3695cab..28e283a  main -> main
```

Default Python: 3.14

argparse error during

ns-3 configuration

Fix: explicitly run system Python 3.12.3

/usr/bin/python3 ./ns3 configure --enable-examples

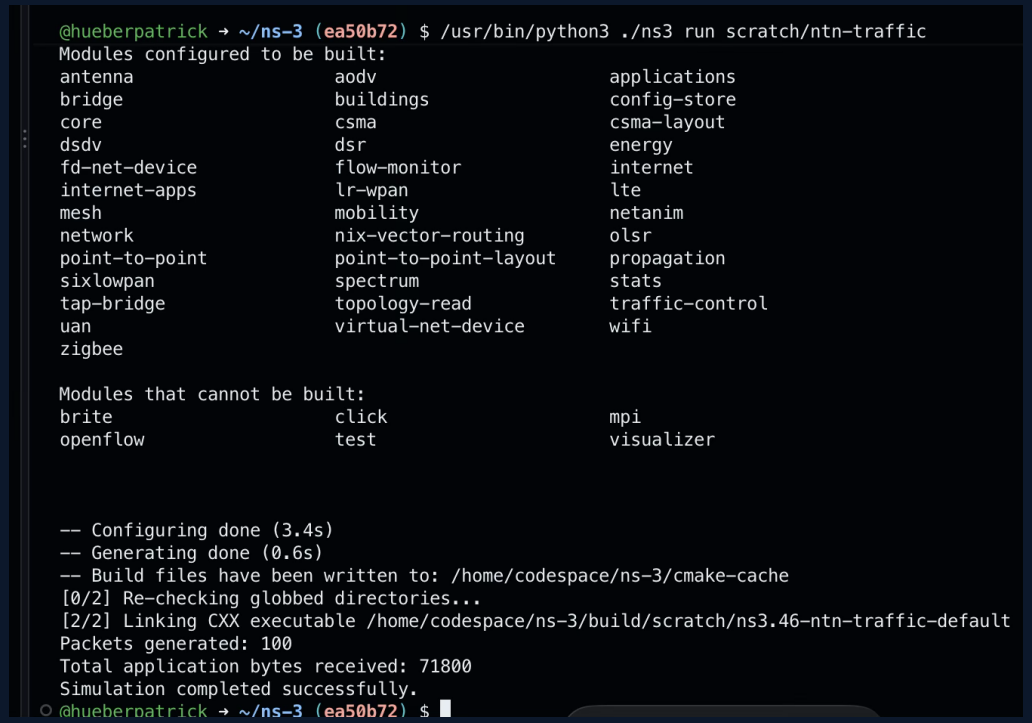
ns-3 built and passed the first functional test

```
@hueberpatrick ➜ /workspaces/MWN-2026-F11502008-hueberpatrick (main) $ cd ~/ns-3
/usr/bin/python3 ./ns3 build -j 2
[0/2] Re-checking globbed directories...
ninja: no work to do.
Finished executing the following commands:
/usr/bin/cmake --build /home/codespace/ns-3/cmake-cache -j 1
@hueberpatrick ➜ ~/ns-3 (ea50b72) $
```

```
/usr/bin/cmake --build /home/codespace/ns-3/cmake-cache -j 1
@hueberpatrick ➜ ~/ns-3 (ea50b72) $ /usr/bin/python3 ./ns3 run first
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 1024 bytes to 10.1.1.2 port 9
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
@hueberpatrick ➜ ~/ns-3 (ea50b72) $
```

Built-in example: 1,024 bytes sent • 3.69 ms one-way • 7.37 ms round-trip

My own traffic generator: 100 UDP packets



- **Two nodes**

  10 Mbps point-to-point link; 10 ms delay.

- **Variable payload**

  200 / 600 / 1,200 bytes, uniformly selected.

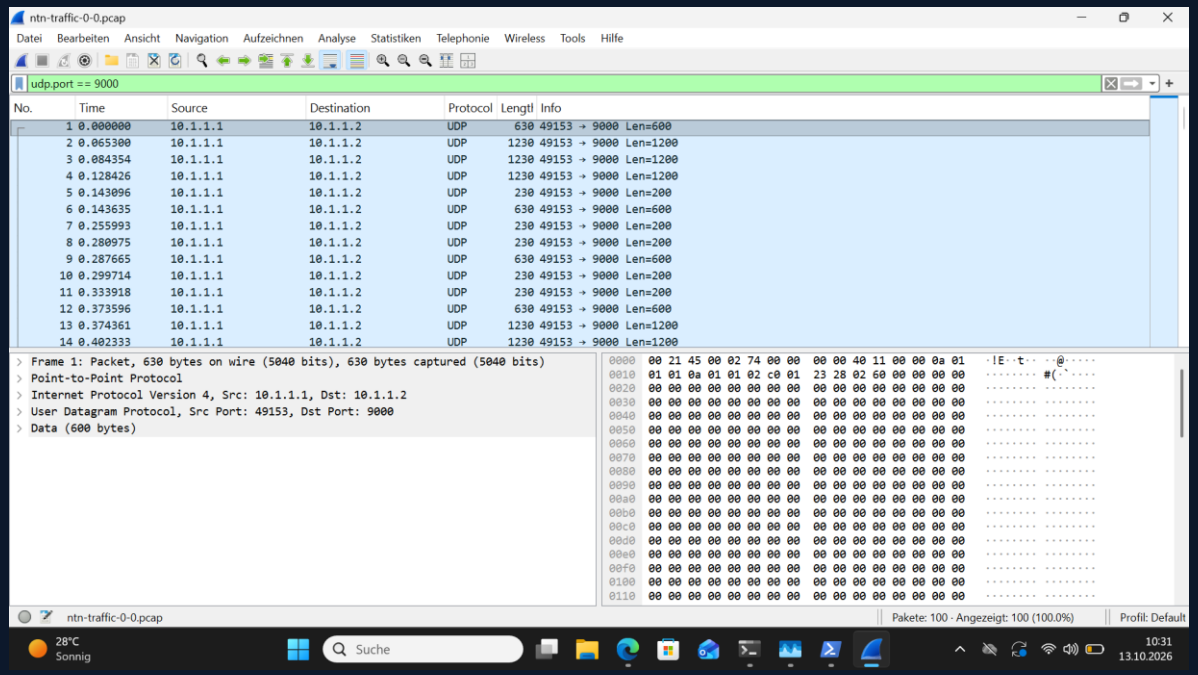
- **Random timing**

  Exponential inter-arrival; mean 50 ms.

- **Successful run**

  100 generated; 71,800 application bytes received.

Verification, lessons learned & next steps



100 / 100 UDP packets visible

CSV + two PCAP files

Packet-size PDF / CDF

Inter-arrival PDF / CDF

Next: validate a 3GPP traffic profile and introduce NTN-specific models + QoS measurements.